
HK IMO Training 2020-2021
Problem Set 16
For 27 February 2021



Problem 61. Find all prime number(s) p such that $p^2 - p - 1$ is the cube of some integer.

Problem 62. Let $p > 2$ be a prime number, $m > 1$ and n be positive integers such that $\frac{m^{pn} - 1}{m^n - 1}$ is a prime number. Show that

$$pn \mid (p-1)^n + 1.$$

Problem 63. Let $a_1, a_2, a_3, \dots$ be an infinite sequence of distinct integers. Prove that there are infinitely many primes p that distinct positive integers i, j, k can be found such that $p \mid a_i a_j a_k - 1$.

Problem 64. For a positive integer n define $S_n = 1! + 2! + \dots + n!$. Prove that there exists an integer n such that S_n has a prime divisor greater than 10^{2012} .